

Project 1 - Logistics Integration & Event Processing

FreightLogics Systems | Contract Software Developer

OVERVIEW

Designed and implemented an event-driven logistics integration workflow connecting internal business systems with external transportation services. The solution enabled reliable shipment processing, fault isolation, and data consistency across distributed systems.

KEY RESPONSIBILITIES

- Developed Azure Functions-based event processors integrated with Azure Service Bus for asynchronous shipment processing.
- Implemented validation controls, duplicate-message detection, retry policies, and dead-letter handling.
- Used Durable Entities to manage fault states and isolate external-system failures.
- Reconciled shipment data in SQL Server and investigated production integration issues.

KEY OUTCOMES

- Improved operational resilience by isolating external API failures.
- Made asynchronous failures traceable through correlation IDs and structured logs.
- Improved consistency between internal records and transportation-service APIs.

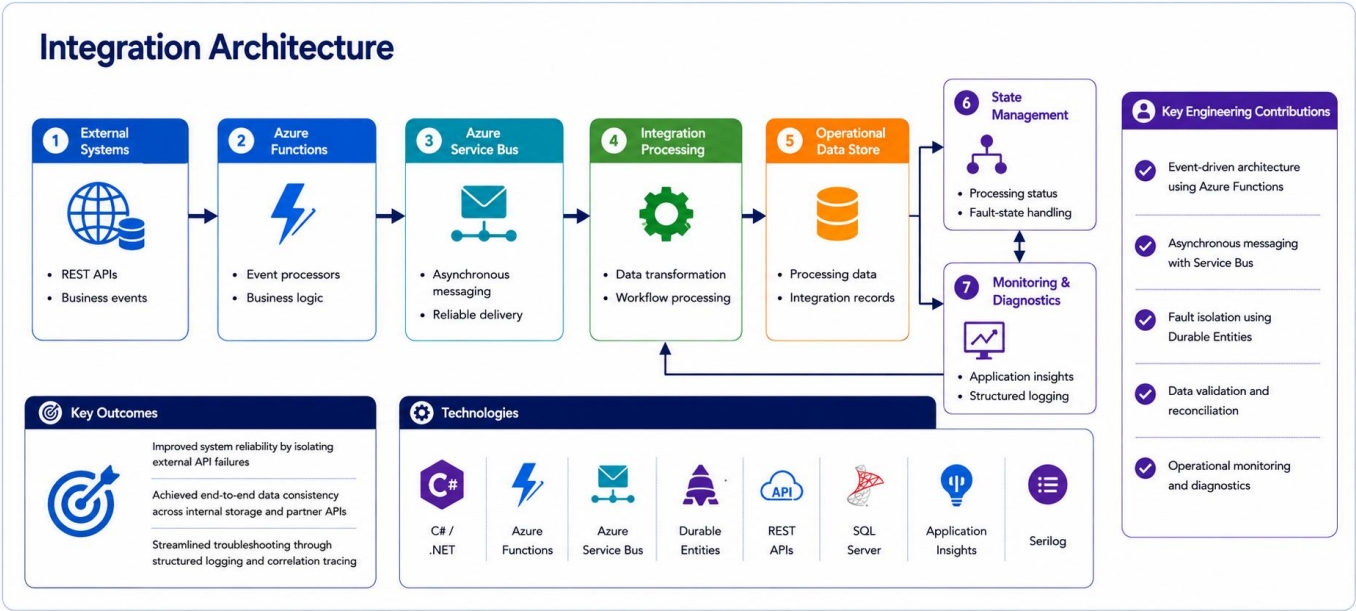
TECHNICAL CHALLENGES

- Maintaining traceability across asynchronous, distributed workflows.
- Preventing partner-system failures from blocking unrelated shipment processing.
- Detecting duplicate or inconsistent data across internal storage and partner APIs.

KEY DECISIONS

- Separated transport messaging from fault-state management to contain failures.
- Used correlation IDs and structured logging for end-to-end investigation.
- Added validation and reconciliation checkpoints instead of relying on successful API responses alone.

TECHNOLOGIES C#, .NET, Azure Functions, Azure Service Bus, Durable Entities, REST APIs, SQL Server, Application Insights, Serilog



Project 2 - Industrial Data Ingestion & Monitoring

Ballard Power Systems | Software Development Engineer

OVERVIEW

Designed and supported a production-grade telemetry ingestion pipeline that collects, validates, and stores high-volume industrial data from distributed FTP and SFTP sources, with automated monitoring and recovery.

KEY RESPONSIBILITIES

- Designed and supported an end-to-end ingestion pipeline for high-volume telemetry data.
- Built validation, missing-data detection, and automated backfill capabilities.
- Investigated production failures using application logs, database records, and processing history.
- Developed monitoring tools that exposed ingestion status and data-quality issues.

KEY OUTCOMES

- Processed approximately 50,000 telemetry files per day.
- Reduced manual monitoring workload by more than 90%.
- Improved detection and recovery of missing and inconsistent records.

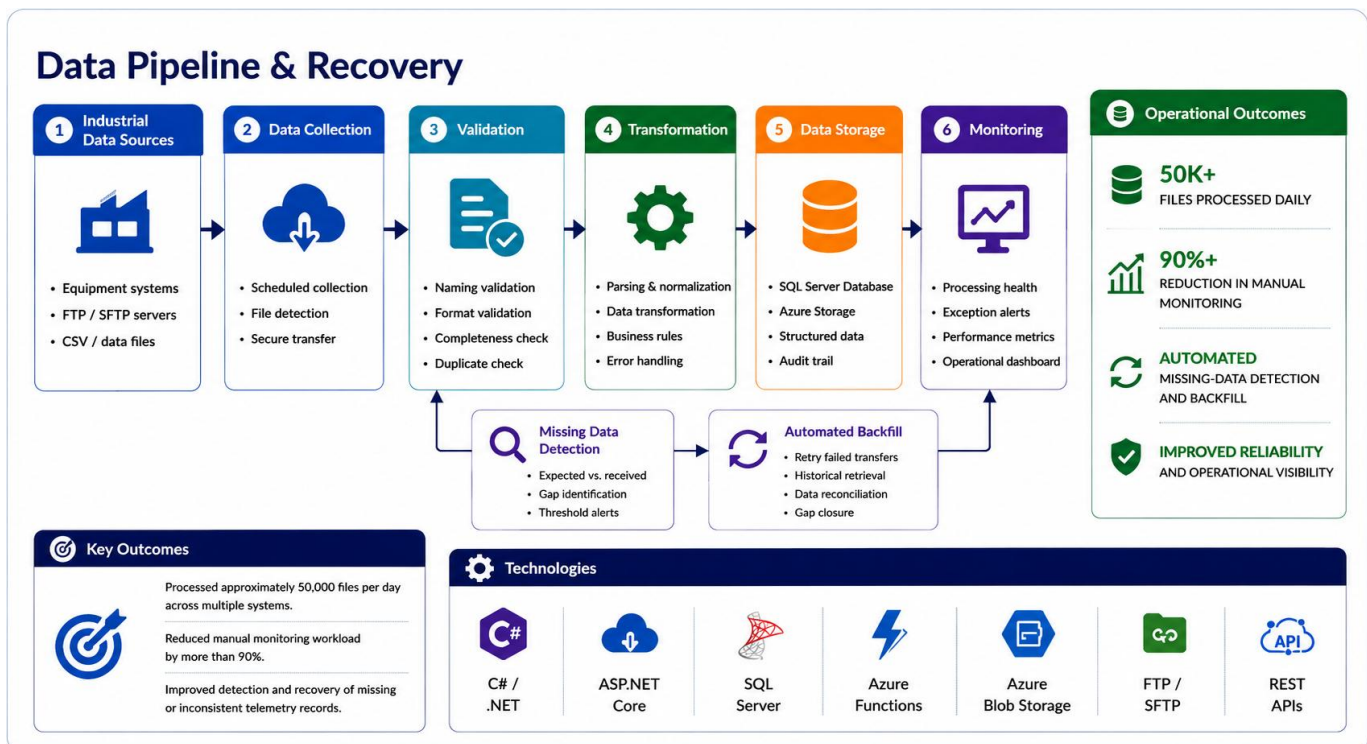
TECHNICAL CHALLENGES

- Processing large daily file volumes from distributed data sources.
- Distinguishing delayed data from genuinely missing or invalid records.
- Recovering failed ingestion without introducing duplicates.

KEY DECISIONS

- Separated ingestion, validation, persistence, and recovery responsibilities.
- Recorded processing state so failed items could be investigated and safely retried.
- Combined automated detection with an operational dashboard for exception-driven support.

TECHNOLOGIES C#, ASP.NET Core, SQL Server, Azure Functions, Azure Blob Storage, FTP/SFTP, REST APIs



Project 3 - Module Assembly Timer

Ballard Power Systems | Production Application

OVERVIEW

Developed and supported a Windows Forms factory-floor application for assembly cycle tracking, process validation, and real-time operator visibility.

KEY RESPONSIBILITIES

- Developed and supported a Windows Forms application used by production operators.
- Implemented real-time timers, process status, pause tracking, and assembly-step validation.
- Reproduced defects on the production line and verified fixes after deployment.
- Improved usability and reliability based on direct operator feedback.

KEY OUTCOMES

- Enabled real-time visibility into assembly progress and build status.
- Supported bottleneck analysis through production-data tracking.
- Improved communication and reliability for production support workflows.

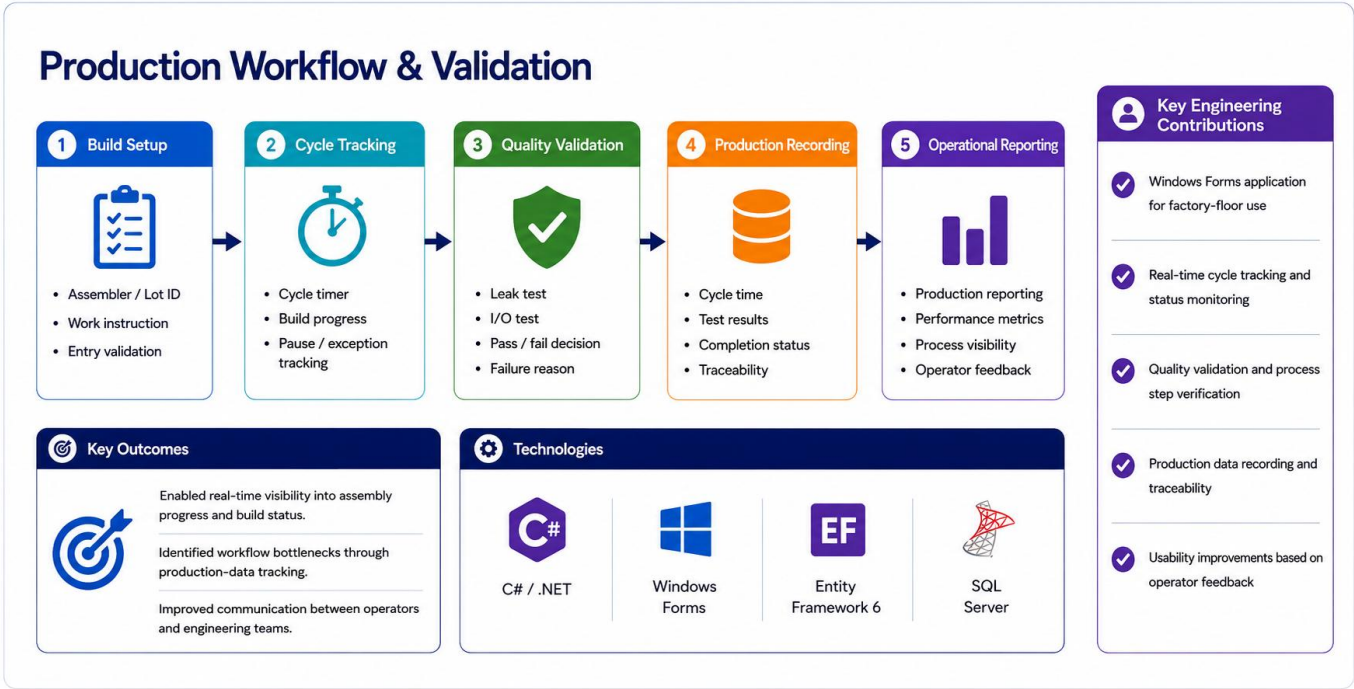
TECHNICAL CHALLENGES

- Keeping the interface clear and responsive for factory-floor use.
- Representing pauses, rework, testing, and partial builds accurately.
- Validating fixes in an active production environment.

KEY DECISIONS

- Used a focused desktop workflow suited to fixed production workstations.
- Captured structured process events instead of relying only on final cycle time.
- Integrated visual status feedback to improve communication between operators and engineering teams.

TECHNOLOGIES C#, .NET, Windows Forms, Entity Framework 6, SQL Server



Project 4 - Azzier CMMS & Mobile

Tero Consulting Ltd. | Software Developer

OVERVIEW

Developed, maintained, and supported configurable enterprise CMMS web and mobile applications for maintenance operations, work orders, asset reliability, and field execution.

KEY RESPONSIBILITIES

- Contributed to CMMS development, maintenance, feature enhancements, testing, and deployment support.
- Implemented customer-requested UI and workflow enhancements using Telerik and Kendo UI.
- Supported REST API, CORS, SQL Server, and mobile data-transfer workflows.
- Investigated customer issues and delivered production fixes and documentation.

KEY OUTCOMES

- Delivered customer-requested enhancements across web and mobile workflows.
- Supported reliable work-order and asset-management operations.
- Improved maintainability through documented fixes and reusable UI customization.

TECHNICAL CHALLENGES

- Supporting customer-specific behavior without destabilizing shared product workflows.
- Maintaining consistent data exchange between web, API, database, and mobile layers.
- Troubleshooting issues across configurable enterprise environments.

KEY DECISIONS

- Kept customer-facing customization within reusable UI and configuration patterns.
- Used API boundaries to separate mobile data transfer from core CMMS business logic.
- Combined development work with deployment validation and production support.

TECHNOLOGIES C#, ASP.NET / .NET Framework, SQL Server, REST APIs, JavaScript, Telerik, Kendo UI, Android, Azure



Project 5 - City Growth Simulation

Independent Project | Data-Driven Simulation

OVERVIEW

Built an interactive urban-planning simulation that compares baseline, projected, and user-defined scenarios through map-based facility placement and measurable city indicators.

KEY RESPONSIBILITIES

- Designed a model that calculates the impact of facility types on multiple urban indicators.
- Implemented scenario-based analysis for baseline, projected, and user-defined outcomes.
- Built an interactive map for placing facilities and visualizing changes within city boundaries.
- Created structured summaries of trends, trade-offs, and policy considerations.

KEY OUTCOMES

- Enabled exploration of how infrastructure decisions influence long-term urban outcomes.
- Made projected trends directly comparable with user-defined scenarios.
- Demonstrated data-driven decision support in an interactive simulation environment.

TECHNICAL CHALLENGES

- Translating qualitative facility impacts into transparent simulation parameters.
- Keeping baseline, projected, and user-defined scenarios comparable.
- Presenting map interactions and indicator changes without overwhelming users.

KEY DECISIONS

- Used explicit deterministic facility-impact rules for explainability.
- Separated baseline, projected, and user-scenario calculations.
- Combined map-based placement with side-by-side outcome summaries.

TECHNOLOGIES React, JavaScript, Leaflet, Data Visualization, Scenario Modeling

